How to Configure Security Permissions

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Configure Security on Sencha Pages

Data Model

The data model for security permissions on Sencha belongs to web.data_security table. It uses indirect referencing to identify on which h object does it needs to apply security permission. For example if we are applying security on a button using Business Function, Entity Id will correspond to Entity Id for business function in Entity table and Key_value would correspond to Id of that Business Function Record. Department_Role_Id belongs to the Department_role the current Security_user belongs to.

```
select * from web.data_security
```

	DATA_SECURITY_ID	ENTITY_ID	KEY_VALUE	PERMISSION_CODE	SECURITY_USERS_ID	DEPARTMENT_ROLE_ID	SECURITY_TAG	INSERTED_BY	INSERTED_DATE
1	7544	5494	2	R		188		CATCLIE	2/20/2013 12:40:20 PM
2	7547	5494	90	R		188		CATCLIE	2/20/2013 12:40:20 PM
3	7548	5494	91	R		188		CATCLIE	2/20/2013 12:40:20 PM
4	11113	5494	1216	R		188		JKLASEN	3/1/2013 2:13:25 PM
5	11196	5494	1218	R		188		FVITI	3/5/2013 10:36:43 PM
6	11210	5494	1221	R		188		IGORDON	3/10/2013 11:36:48 AM

UI Configurations

There are two ways security can be configured through UI:

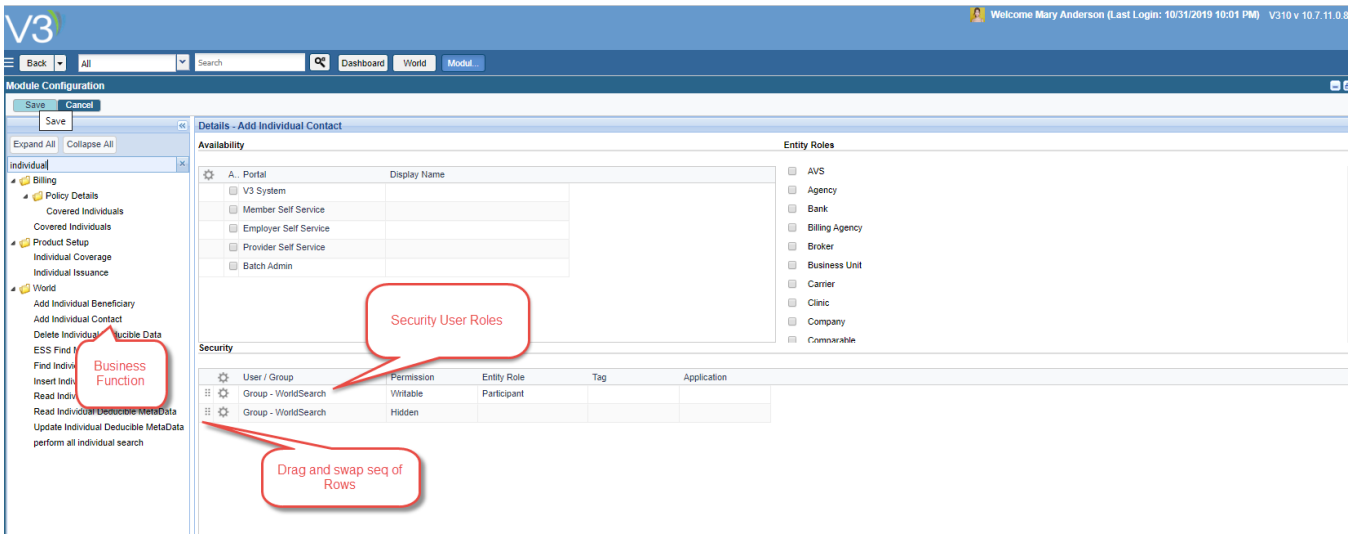
Object Inspector

To understand how to use Object Inspector to configure security permissions, please check the link **Object Inspector .**

Module Configuration

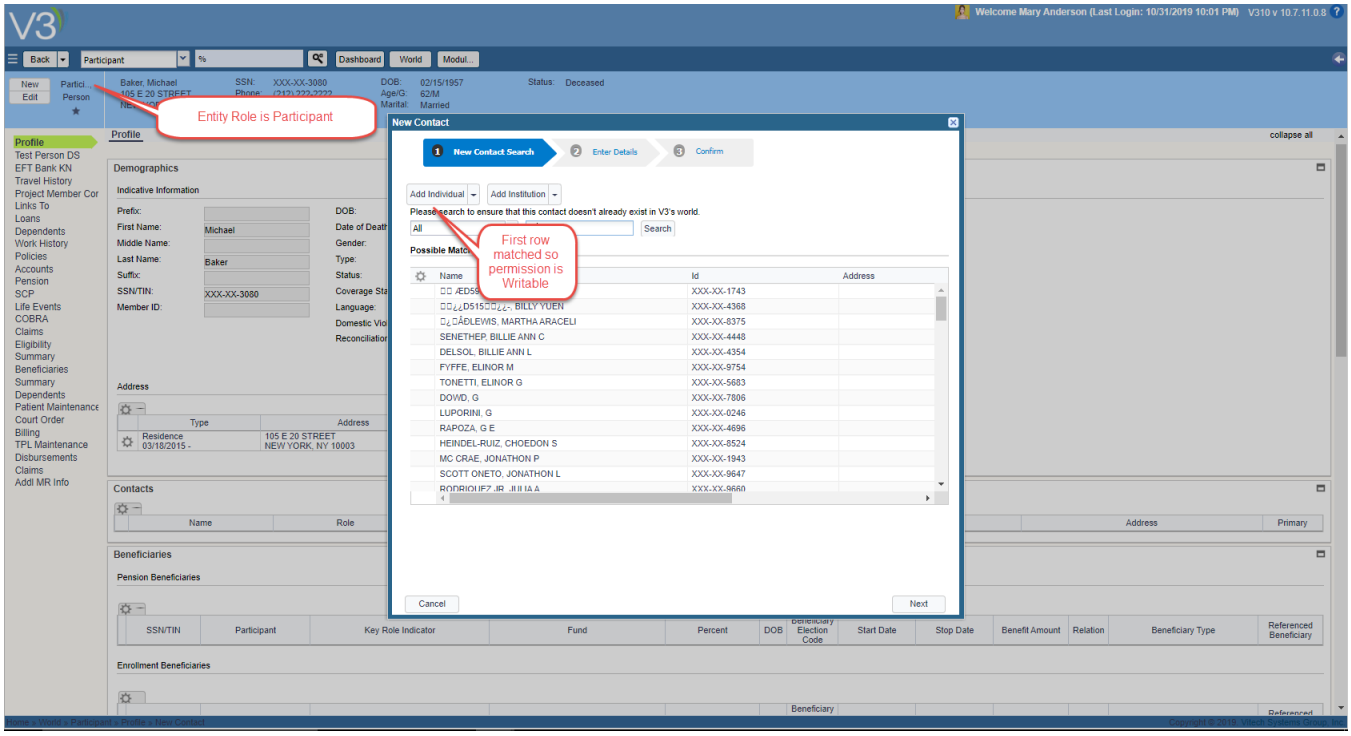
Business Function Configuration Under Modules

Go to Hamburger on top left in V3Application. RuleModule Configuration

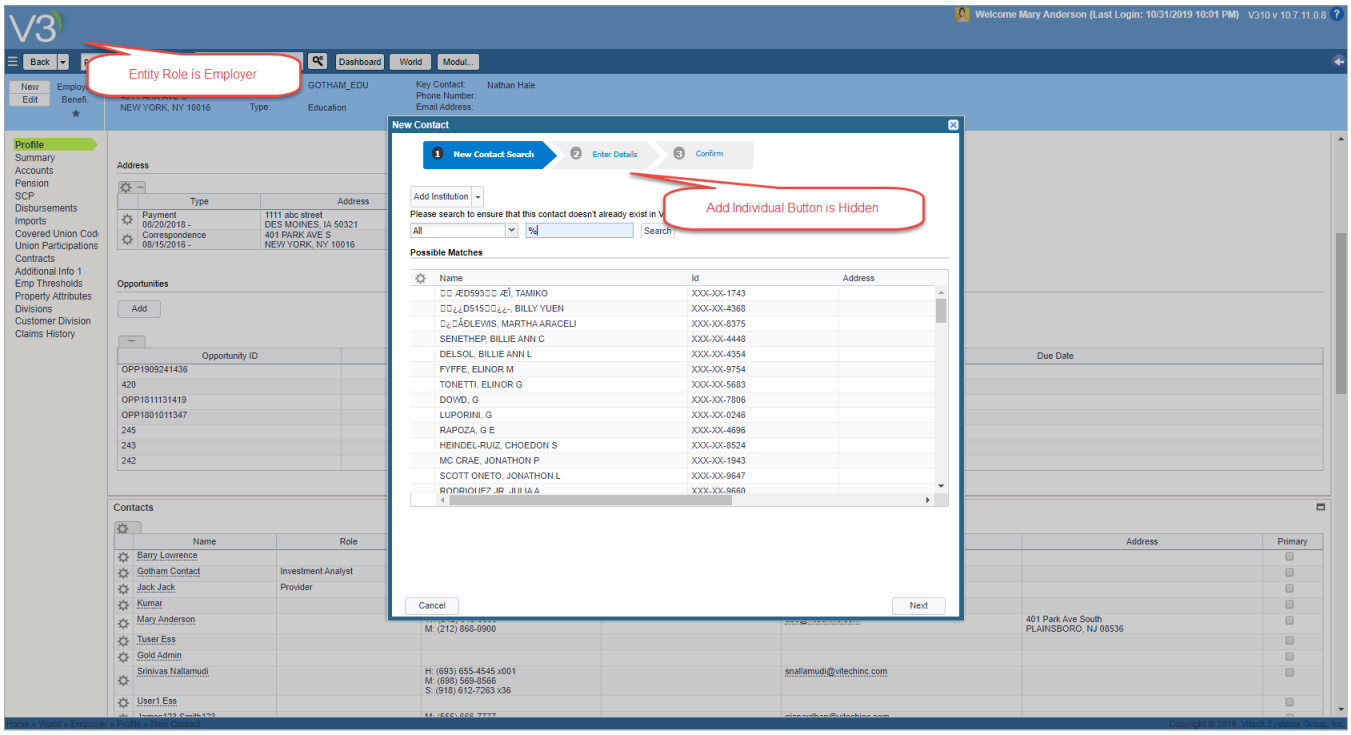


In the above screen, left Pane folders are the Modules. The Links below them represent Business Functions. Once we select one the business functions, the window will show a grid named Security. This grid can be used to configure security for the business function and module. The grid is scanned from top to bottom and the first row that matches all the conditions is picked up for security permission.

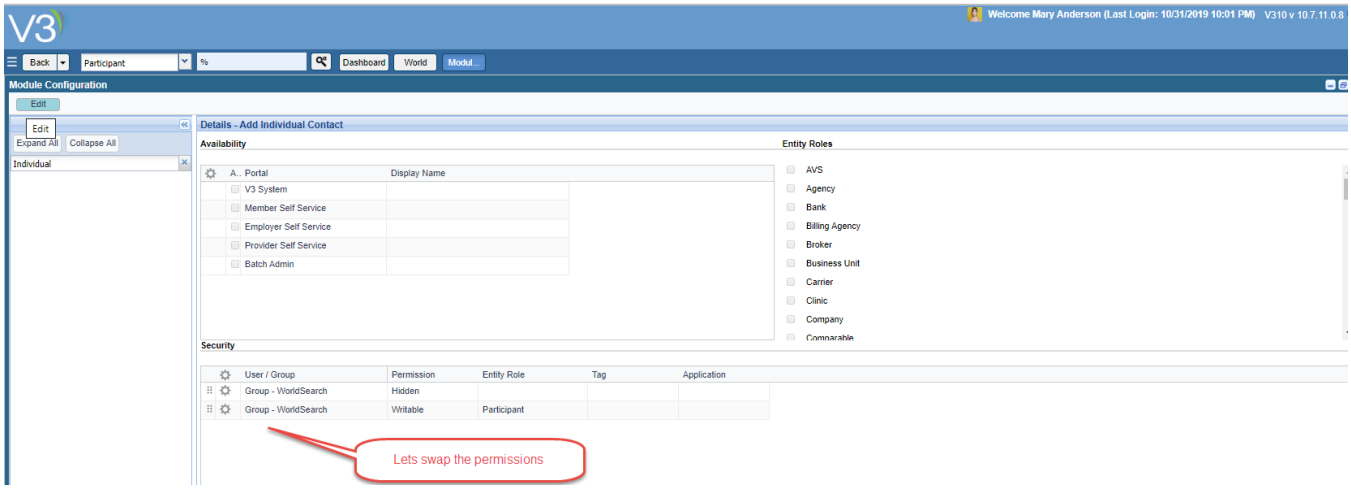
In order to understand how does the configurations in this grid work, lets go over the example configuration above. For the same security user role, there are two permissions configured. One is W and next is H. So if logged in user have Department Role Group-WorldSearch assigned to it, it will check what is the entity role. If the entity role is participant, First row would match and Permission would be writable as seen below:



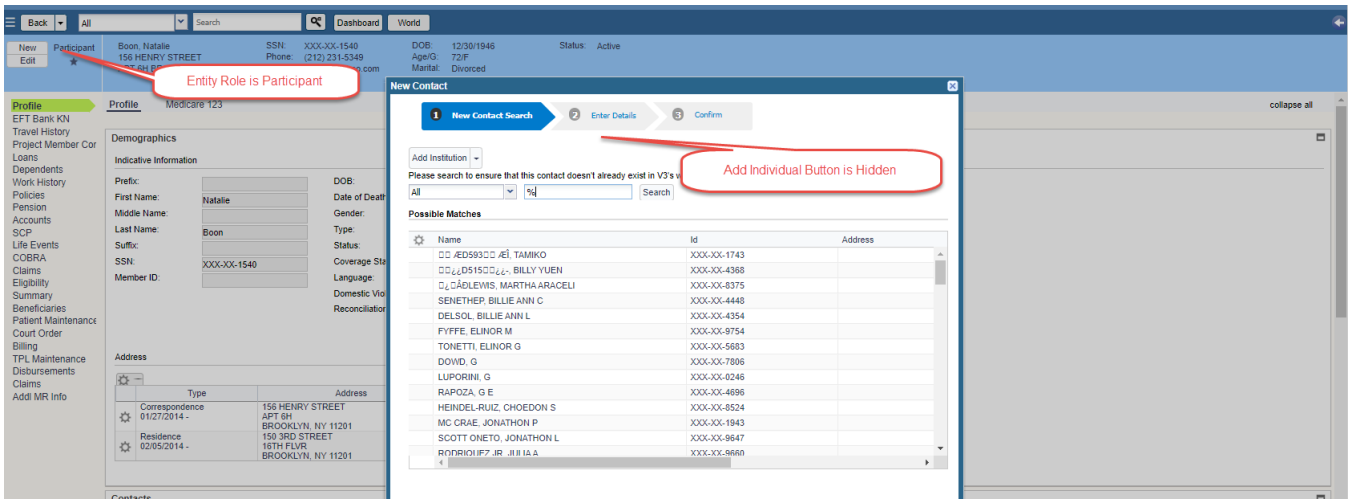
In example below, if an Employer is loaded, The first row in Security Grid would not match because entity loaded is an employer and not a participant. So first row security Permission would not get applied. The next row(2nd) will match because role is not specified so it would be applicable for all row. Since second row matched, the applied permission for Emplyer is Hidden and Add Contact Button would be Hidden.



Lets swap the order of permissions now:



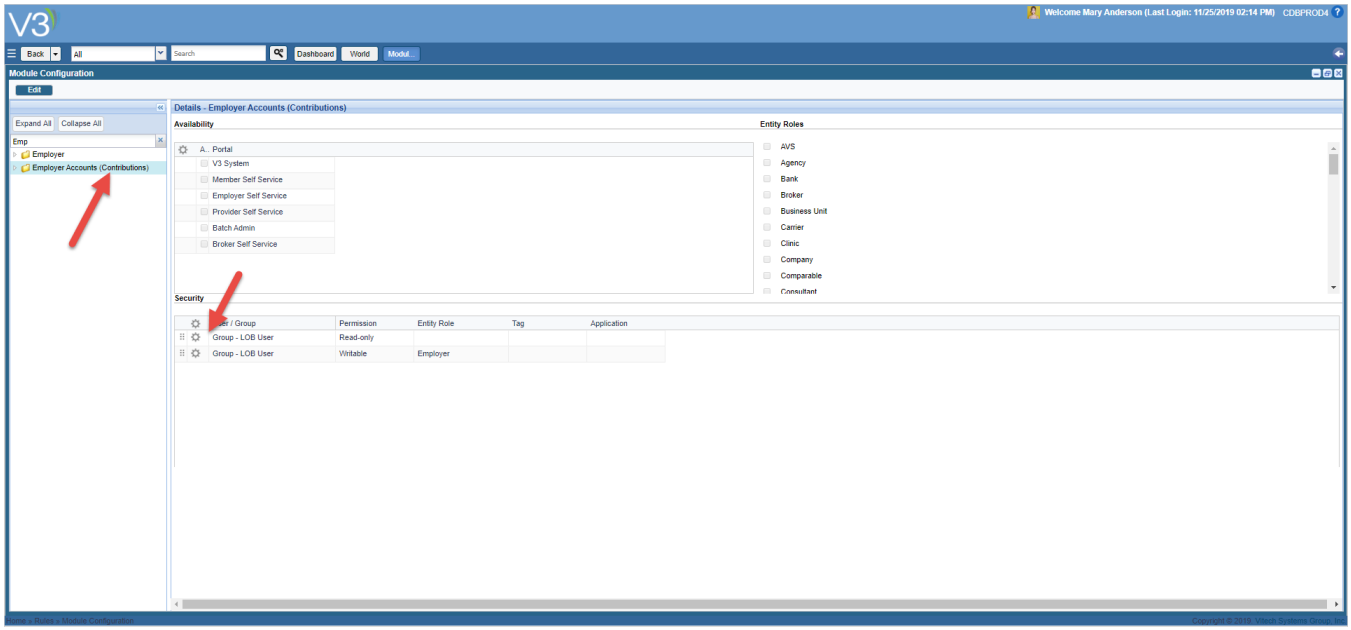
Now if we load participant, even though both rows match , first row would match first and the permission applied would be hidden



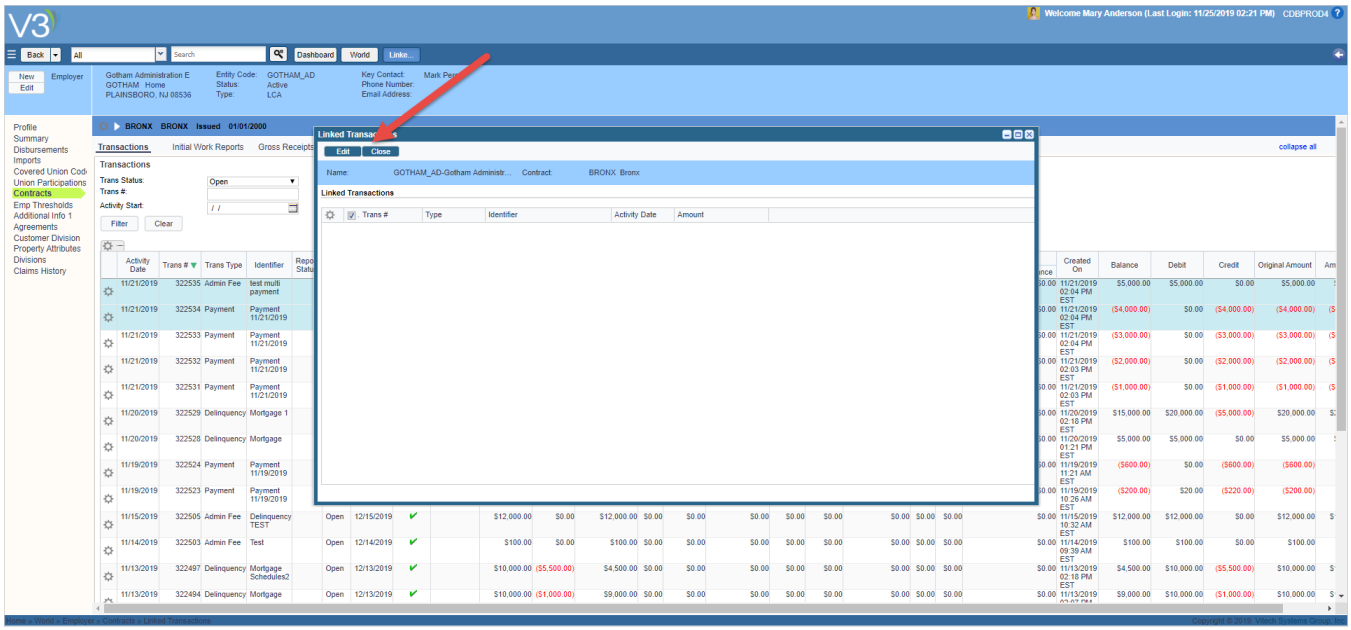
If the user has two Security Roles then like above examples, first matching row would be applied. For example if user has LOB_USER Role and WorldSearch Role and all other columns are same in two records then first matching row would get applied.

Applying Security to whole Sencha Pop Up using Module Security

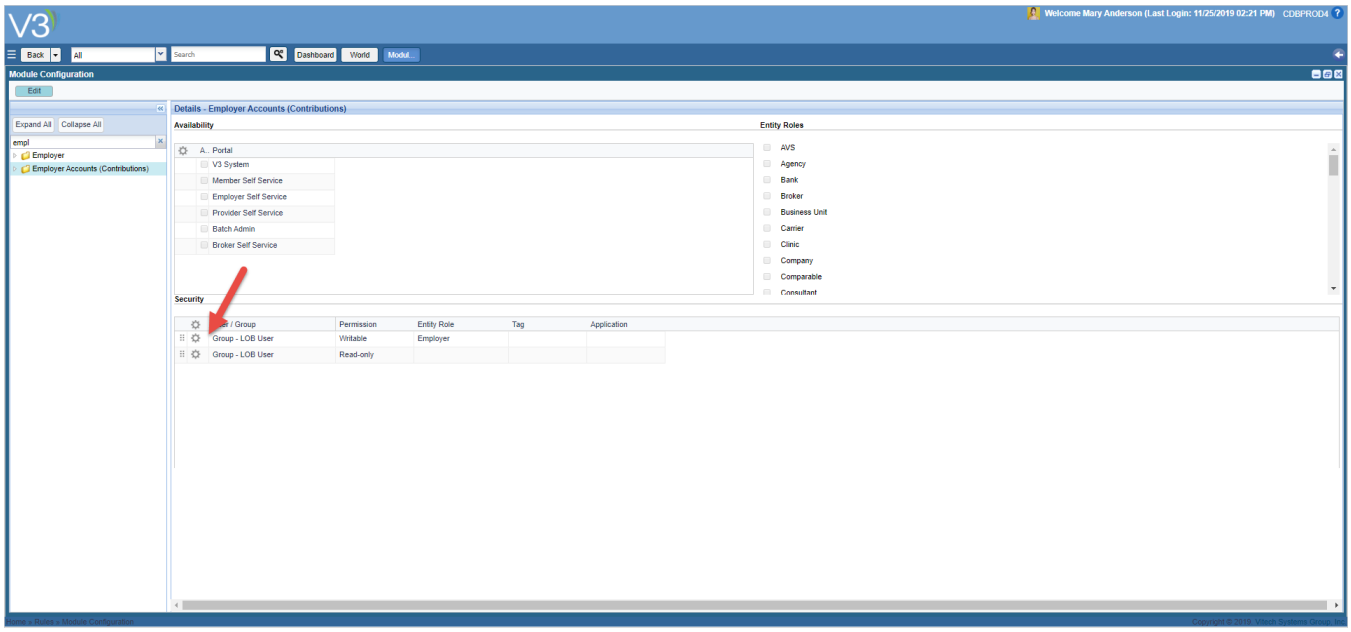
We can secure Sencha Popups and control which mode they will open using Module Permissions. It works the same way as business functions and uses the same data model (Data_security) table. This time the entity Id will be the Modules record Id in Entity Table and key value will be the module Id of that module.



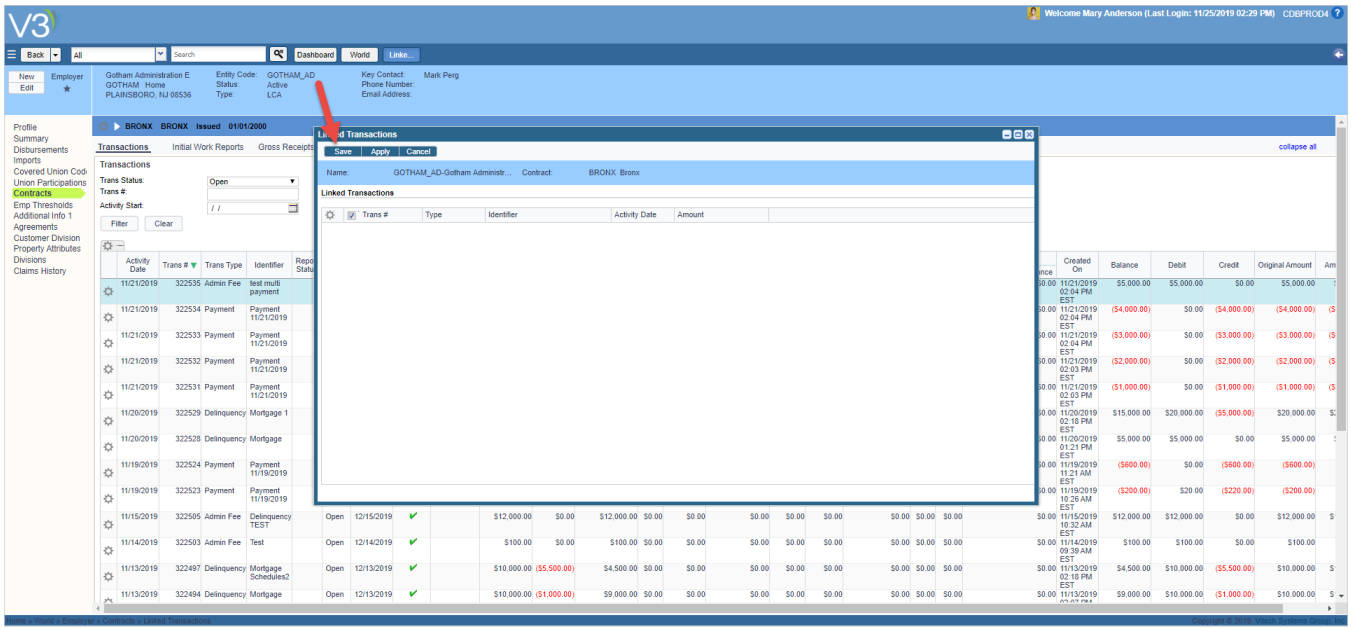
As we can see the pop up below opened in read only mode because the security with lowest seq(first row in above grid) was readonly and it was applicable to the scenario below.



Now if we swap the security to have the Writable permission first in the grid(lowest seq number)



As we can see the writable permission is applicable now the grid is opening up in writable mode.



Precedence of Data Security records.

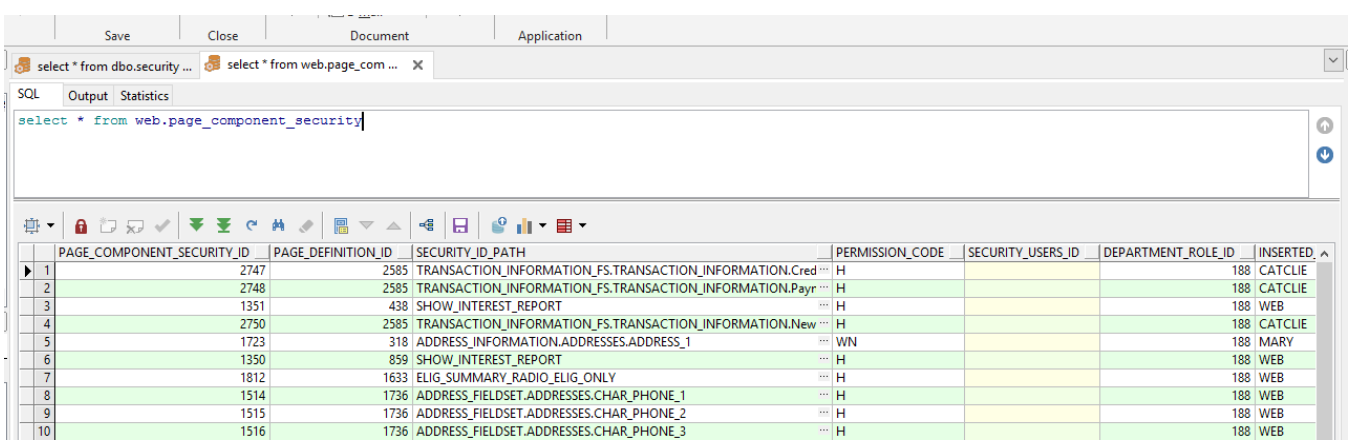
This section describes the order priority of DataSecurity and which one takes precedence.

- Lets say we have a user: MARY
- MARY has 2 roles:
  - ROLE_1
  - ROLE_2
- For some UI element (lets say it is an EntityAttribute), I have 2 conflicting DataSecurity configuration records.
  - EntityId/KeyValue/BusinessEntitySubtype/Application are all identical in both DataSecurity records.
  - One record says the permission is W and the other says it is H.
- Which DataSecurity record will be applied on my UI element?
  - The answer for that lays in the SEQ_NO of DataSecurity record. The one with the lower SEQ_NO will take precedence.
  - Please note that if both SEQ_NO are identical (or missing), we will not guarantee the precedence of one record over the other, V3 will just pick a record.
  - It is advisable while configuring V3 to avoid such a scenario by populating the SEQ_NO field.

Configure Security on Tapestry Pages

Data Model

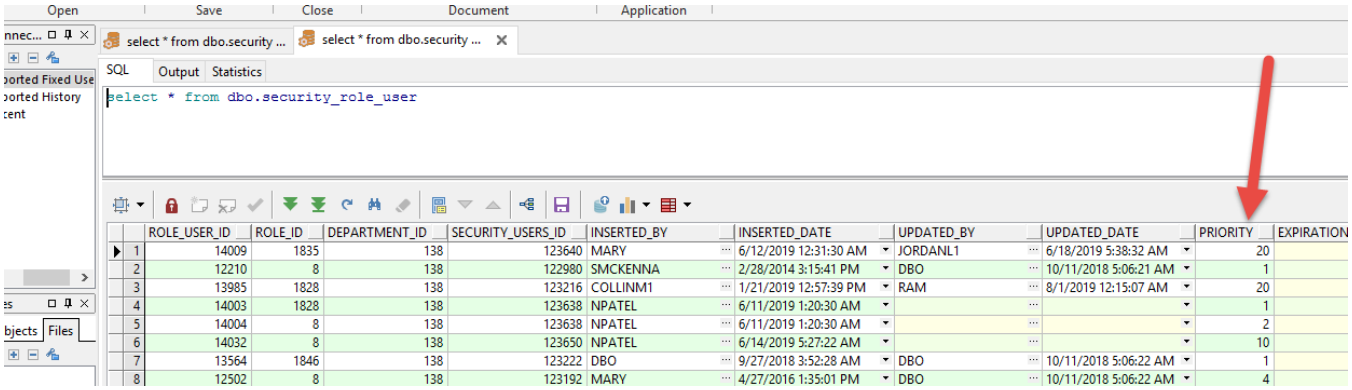
The Security Permissions on Tapestry Pages correspond to web.page_component_security table. The components on Tapestry pages are assigned a security path id. We can register the security permissions against the Page and Security Id Path combination.



	PAGE_COMPONENT_SECURITY_ID	PAGE_DEFINITION_ID	SECURITY_ID_PATH	PERMISSION_CODE	SECURITY_USERS_ID	DEPARTMENT_ROLE_ID	INSERTED_
1	2747	2585	TRANSACTION_INFORMATION_FS.TRANSACTION_INFORMATION.Cred	H		188	CATCLIE
2	2748	2585	TRANSACTION_INFORMATION_FS.TRANSACTION_INFORMATION.Payr	H		188	CATCLIE
3	1351	438	SHOW_INTEREST_REPORT	H		188	WEB
4	2750	2585	TRANSACTION_INFORMATION_FS.TRANSACTION_INFORMATION.New	H		188	CATCLIE
5	1723	318	ADDRESS_INFORMATION.ADDRESSES.ADDRESS_1	WN		188	MARY
6	1350	859	SHOW_INTEREST_REPORT	H		188	WEB
7	1812	1633	ELIG_SUMMARY_RADIO_ELIG_ONLY	H		188	WEB
8	1514	1736	ADDRESS_FIELDSET.ADDRESSES.CHAR_PHONE_1	H		188	WEB
9	1515	1736	ADDRESS_FIELDSET.ADDRESSES.CHAR_PHONE_2	H		188	WEB
10	1516	1736	ADDRESS_FIELDSET.ADDRESSES.CHAR_PHONE_3	H		188	WEB

How to configure different security permissions for different roles and how to know which one would take precedence

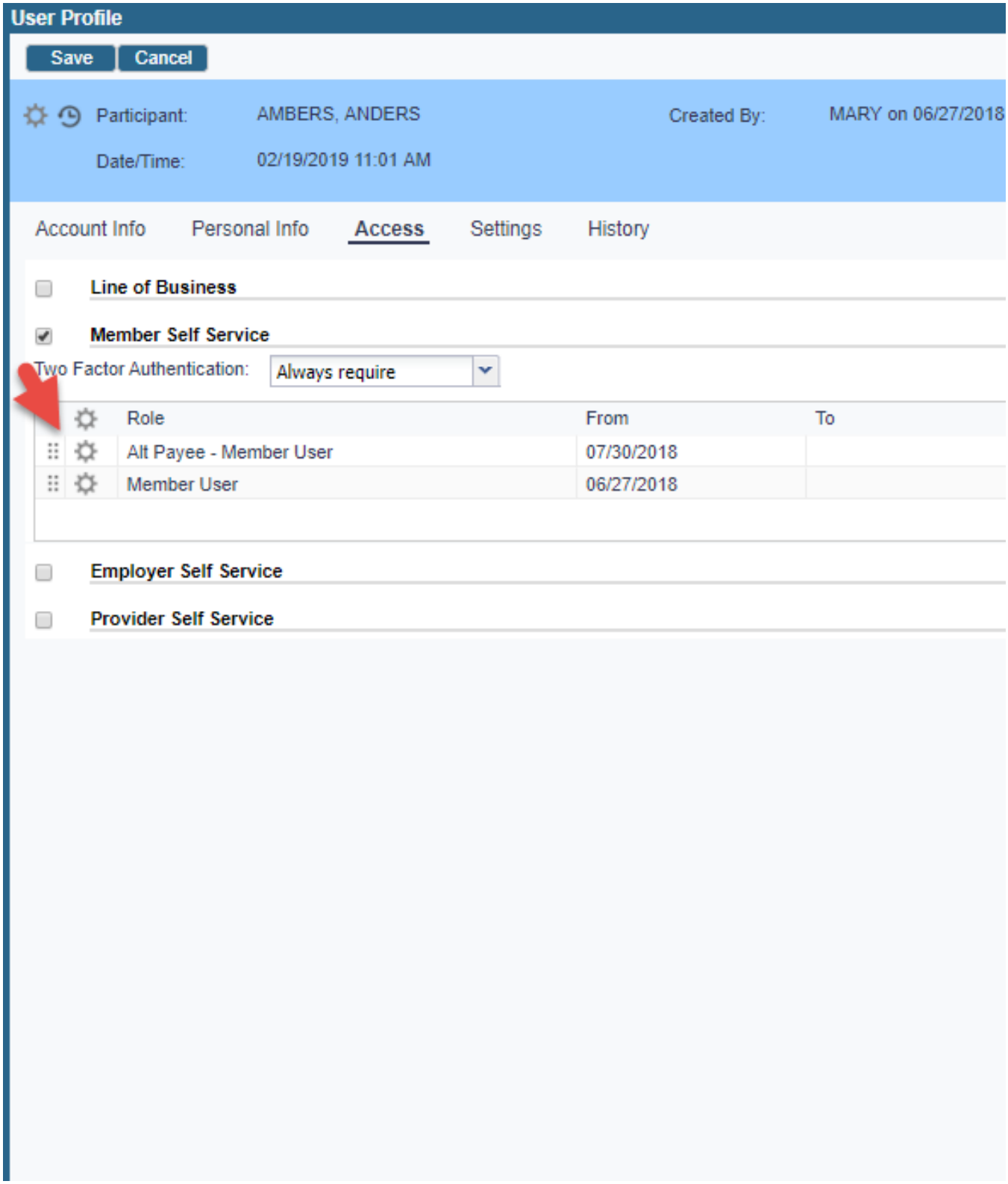
A given security user can be associate to multiple department roles. Database table Security_Role_user controls this binding. This table has a column called Priority. This table tells which role of a security user will take priority on deciding which security permission would apply. For example if a user MARY is associated with two roles, LOB USER and ADMIN, and ADMIN has more priority then if two records are configured in page_component_security table for same user + same Page definition_ same security id path then the one which has role ADMIN would get applied.



	ROLE_USER_ID	ROLE_ID	DEPARTMENT_ID	SECURITY_USERS_ID	INSERTED_BY	INSERTED_DATE	UPDATED_BY	UPDATED_DATE	PRIORITY	EXPIRATION
1	14009	1835	138	123640	MARY	6/12/2019 12:31:30 AM	JORDANL1	6/18/2019 5:38:32 AM	20	
2	12210	8	138	122980	SMCKENNA	2/28/2014 3:15:41 PM	DBO	10/11/2018 5:06:21 AM	1	
3	13985	1828	138	123216	COLLINM1	1/21/2019 12:57:39 PM	RAM	8/1/2019 12:15:07 AM	20	
4	14003	1828	138	123638	NPATEL	6/11/2019 1:20:30 AM			1	
5	14004	8	138	123638	NPATEL	6/11/2019 1:20:30 AM			2	
6	14032	8	138	123650	NPATEL	6/14/2019 5:27:22 AM			10	
7	13564	1846	138	123222	DBO	9/27/2018 3:52:28 AM	DBO	10/11/2018 5:06:22 AM	1	
8	12502	8	138	123192	MARY	4/27/2016 1:35:01 PM	DBO	10/11/2018 5:06:22 AM	4	

Configuring Priority for security roles for a user from UI

The priority of which role will take precedence can be configured from UI. Go to Hamburger Admin User Maintenance Details Access. You can swap the order of Roles for each application. This will decide the priority of roles. So in example below, security permission which has role Alt Payee will take precedence in MSS.



Precedence between Security Roles and Security Tags

In example above we saw which role will take precedence if a security permission record is configured for two roles. Now lets see what happens if security permission is configured for two records with one having security tag and one with Security Role Priority. To clarify, priority always takes precedence over security tags.

IF there are two records, one with security tag and one without security tags, the one with security tag would take precedence.

To summarize above points we can look at the table below:

					Mary		
On UI, Proprity is controlled from the User profile screen role order						Role	
						2	LOB
						3	Search
Assume for same page definition + security id path + security user, we have multiple permissions configured						1	Admin
Case 1	Permission	Tag	Security_ROLE_USER.Priority	Role name			
1	W		1	Admin			
2	R	ABC	2	LOB			
3	H		2	LOB			
	Applied Permission = W since Priority will take precedence						
Case 2							
	Permission	Tag	Security_ROLE_USER.Priority				
1	W	ABC	2				
2	H		2				
	Applied Permission = W since Tag will take precedence over non tag records						

As we can see in Case 1, record number 1 has highest priority so applied permission would be W. In case 2, both records have same priority but since record 1 has security tag, applied permission would be W

The V3Button special case

While being a Tapestry component, the V3Button could optionally define a BusinessFunction (jwc parameter named: "businessFunction"). When that happens, the permission on the V3Button will behave as if it was a Sencha button and will completely bypass Tapestry security. Please refer to the Sencha security section for details on how does Sencha security functions.

Navigation Security

Navigation security is integral to all the navigation we have in V3. Not just navigations, it impacts both Module and Business Function Securities as explained below.

Data Model for Navigation

The navigation are tied to web.navigation table.

Navigation security is tied to web.navigation_security table which refers to navigations using navigation_id column.

How navigation permissions are retrieved:

Navigations are considered as a tree and permissions are fetched from leaf to root order. For example If there are three Navigations A, B, C and A is parent of B and B is parent navigation of C then below is the pseudo code to get permission for C

check if C is inactive if yes return Hidden

else check if parent of C i.e. B is inactive if yes return hidden

else check if parent of B i.e. A is inactive if yes return hidden

if none of the above is inactive then actually check permission code for c, if permission is there return permission

else check permission for parent of C i,e, return permission for B

else check permission for parent of B i.e. return permission of C

the active_flag and parent_navigation ids are checked in web.navigation table.

How navigation security affects module and business function permissions

Module Permissions:

Whenever module permissions are fetched for a module, below is the logic how navigation permissions affect it:

1. For module X, check the navigation table and see which navigation is attached to it - select n.navigation_id from web.navigation where module_id = X
2. If Navigation is there then permissions will be fetched for this navigation based on Navigation permission section above.
3. If navigation permissions are present then return navigation permission, otherwise return module permission from data_security table
4. If none of the permissions are present then get the parent module of module X, say Y and repeat steps 1 to 4

Business Function Permissions

1. For Business Function BF, get the permissions from data_security table.
2. If permissions are present, return permissions
3. Else return module permissions explained above for this Business Function's module

Example explaining above cases

Lets say we have a Business Function named BF. Suppose the Module_id in Business_function table points to module X. Lets also say module X is attached to Navigation C in web.navigation table and C has parent navigation as B

Lets consider some permissions and effective permissions here

Case 1.

BF - Business Function - Permission Code(H)

X - Module - Permission Code(R)

C - Navigation - Permission Code(W)

Effective permission = H

Explanation - Since while getting Business Function permission, it will return permission H, the logic will return H permission immediately without going to check for module permissions

Case 2.

BF - Business Function - No permission

X - Module - Permission Code(R)

C - Navigation - Permission Code(H)

Effective permission = H

Explanation - Since Business Function permission is null, it will return go to check for module permissions. While getting module permission, it will first check for navigation permission and since it will return H, it will immediately return H permission without checking module permission R

Case 3.

BF - Business Function - No permission

X - Module - Permission Code(R)

C - Navigation - No Permission

Effective permission = R

Explanation - Since Business Function permission is null, it will return go to check for module permissions. While getting module permission, it will first check for navigation permission and since it will return null, it will now check for module permission and return R

Case 4.

BF - Business Function - No permission

X - Module - Permission Code(R)

C - Navigation - No permission

B - Navigation - Permission Code(W)

Effective permission = W

Explanation - Since Business Function permission is null, it will return go to check for module permissions. While getting module permission, it will first check for navigation permission and since it will return null, it will check parent navigation of C i.e. B. This will return W permission code and immediately return without checking module permission R

In summary, Business Function permissions, if present, overrides module and navigation permissions. If they are not present then Navigation permission (and all the parent navigation permissions) wins over the module permission. In the end module permissions are returned

Security Permissions on Navigation Records

While Navigation records can hold the R permission, in reality the R permission doesn't do anything (R is the same as W). This is because by the nature of a Menu Item, clicking on a Menu Item do not actually perform a W operation (we are only opening a popup / performing navigation - neither of those actions are considered as a W operation). As such, only H permissions are meaningful for Menu Items. If the R behavior is desired, the R permission has to be applied on the page/popup the Menu Item navigates to.

Permission Propagation between different Permission Types (Tapestry)

The lists below describes how permissions will propagate from one security type to another. Please note that the permission propagation is different based on the type of permission. For the sake of permission propagation, we recognize 2 types of permissions.

Accessibility Permission

This permission is affecting if a UI element is accessible (can we trigger the action on the UI element). A R permission would be Accessibility Permission (R means read only, so buttons are not clickable).

Visibility Permission

This permission is affecting if a UI element is visible (can we see the UI element). A H permission would be Visibility Permission (H means hidden. So UI Element is not visible).

Propagation

For the lists below, the lower the number the higher the priority. So 1 is higher priority than 2, etc...

For Accessibility Permissions, permissions propagate as defined below.

1. PageComponent Security
2. Page Security
3. Navigation Security
4. Module Security

For Visibility Permissions, permissions propagate as defined below.

1. Module Security
2. Navigation Security
3. Page Security
4. Page Component Security

Why is there different Propagation priority between Accessibility and Visibility?

Accessibility

- When one marks a higher level priority element as inaccessible (Read Only). Then by definition all of the child elements will be inaccessible (Read Only) as well. An example for that is marking a Page read only, than all fields within the page will also be Read Only.
- However for the cases where we want to make all fields on a page as Read Only, save for 3 buttons. We wanted to be able to mark the Page as Read Only, and 3 Buttons as Writeable.
- The alternative for that is marking every field in the page as Read Only save for the 3 that need to be writeable.
    - This leads to much more configuration required to achieve the same.

Visibility

- When one marks a higher level priority element as invisible (Hidden). Then by definition, we are no longer able to see any of the child elements.
- Hiding a Page means the page is no longer visible, meaning all the fields that belong to the page are also not visible.

Submitting a JIRA about when the rules above fails.

Tapestry

When it comes to reporting issue with the Permissions mechanism in Tapestry. The Security Team will only accept JIRAs if they follow the template as illustrated by: (VTHREE-92458). Please follow the steps below in order to create the JIRA.

- Open VTHREE-92458 - Getting issue details... STATUS and copy the description to your JIRA
  - To efficiently copy the template to your JIRA. While editing the JIRA details. On the bottom of the editable box, please click on the Text tab. Copy the text from the Template JIRA and place it to your own JIRA. Then click on the Visual tab to return to normal.
- Please populate the **Query Values** section.
- Please populate the **Issue Description** section describing what is not working property and is in conflict with the description of field security should work (described in this page).
- Please populate the **Technical Details** section with additional information about technical steps taken to reproduce the issue. This section can include Breakpoint Screenshots or Step description on how to reproduce the issue.
- Please execute the following query on your DB instance (Note that you have to populate the SecurityUserId, SecurityIdPath and PageDefinition InternalName values). Your populated query belongs in the **Query** section within the JIRA.

Template Script for Tapestry Security

```
with
sa as
(
        select sa.application_id from dbo.security_application sa where sa.internal_name = ?
),
su as
(
        select su.security_users_id from dbo.security_users su where su.user_name = ?
),
pd as
(
        select pd.page_definition_id, pd.internal_name from web.page_definition pd where pd.internal_name = ?
),
pcs as
(
        select
                pcs.page_component_security_id, pcs.page_definition_id, pcs.security_id_path, pcs.
permission_code, pcs.security_users_id, pcs.department_role_id, pcs.security_tag, pcs.
business_entity_subtype_id, pd.internal_name as page_name
        from
                web.page_component_security pcs,
                pd
        where
                pcs.page_definition_id = pd.page_definition_id
),
pcs_role as
(
        select
                pcs.page_component_security_id,
                pcs.page_name,
                pcs.security_id_path,
                sru.priority,
                pcs.business_entity_subtype_id,
                pcs.security_tag,
                pcs.permission_code
        from
                pcs,
                dbo.security_department_role sdr,
                dbo.security_role_user sru,
                dbo.security_role sr,
                sa,
                su
        where
                pcs.department_role_id = sdr.department_role_id
                and sdr.department_id = sru.department_id
                and sdr.role_id = sru.role_id
                and sru.role_id = sr.role_id
                and sr.security_application_id = sa.application_id
                and sru.security_users_id = su.security_users_id
```

```
                and sdr.department_role_id not in(-1)
                and (select dbo.f_is_time_expired(sru.expiration_time) from dual) = 'N'
),
pcs_su as
(
        select
                pcs.page_component_security_id,
                pcs.page_name,
                pcs.security_id_path,
                -1 as priority,
                pcs.business_entity_subtype_id,
                pcs.security_tag,
                pcs.permission_code
        from
                pcs,
                su
        where
                pcs.security_users_id = su.security_users_id
),
combined_pcs as
(
        select * from pcs_su
        union
        select * from pcs_role
)
select *
from combined_pcs
where combined_pcs.security_id_path = ?
-- if SystemSetting PCS_PATH_FIRST = Y use this order by clause
/*
order by
        combined_pcs.security_id_path desc nulls last,
        combined_pcs.PRIORITY,
        combined_pcs.business_entity_subtype_id desc nulls last,
        combined_pcs.SECURITY_TAG desc nulls last
*/
-- if SystemSetting PCS_PATH_FIRST = N use this order by clause
/*
order by
        combined_pcs.PRIORITY,
        combined_pcs.security_id_path desc nulls last,
        combined_pcs.business_entity_subtype_id desc nulls last,
        combined_pcs.SECURITY_TAG desc nulls last
*/;
```

- Take a screenshot of the results and attach in the **Query Results** section of your JIRA.

Sencha

For Sencha, please send JIRAs that follow the template as illustrated by: VTHREE-92738. Please follow the steps below in order to create the JIRA.

- Open VTHREE-92738 - Getting issue details... STATUS and copy the description to your JIRA
  - To efficiently copy the template to your JIRA. While editing the JIRA details. On the bottom of the editable box, please click on the Text tab. Copy the text from the Template JIRA and place it to your own JIRA. Then click on the Visual tab to return to normal.
- Please populate the **Query Values** section which should include the fields to populate the query like security user id, entity id, key value used for data security, security tags, security user role etc
- Please populate the **Issue Description** section describing what is not working property and is in conflict with the description of field security should work (described in this page).
- Please populate **Technical Details** section with additional information about technical steps taken to reproduce the issue. This section can include Breakpoint Screenshots or Step description on how to reproduce the issue.
- Please execute the following query on your DB instance (Note that you have to fill the query values in place of question marks) and attach the screenshot of results. Your populated query belongs in the **Query** section within the JIRA.

Template Script for Sencha Security

```
with
  security_application_user_set as
  (
    select
```

```
      sau.application_user_id,
      sau.application_id,
      sau.security_users_id
  from
      dbo.security_application_user sau,
      dbo.security_application sa,
      dbo.security_users su
  where
      sa.internal_name = ?
      and su.user_name = ?
      and sau.application_id = sa.application_id
      and sau.security_users_id = su.security_users_id
      and sau.activated_flag = 'Y'
      and sau.suspended_flag = 'N'
      and su.account_active = 'Y'
      and su.account_locked_flag = 'N'
      and sa.active_flag = 'Y'
)
select
    ds.entity_id,
    ds.key_value,
    ds.DEPARTMENT_ROLE_ID,
    ds.seq_no,
    ds.business_entity_subtype_id,
    ds.security_tag,
    ds.permission_code,
    ds.application_id
from
    web.data_security ds,
    security_application_user_set
where
(
    ds.APPLICATION_ID = security_application_user_set.application_id or ds.APPLICATION_ID is null
)
and
(
    exists
    (
        select
            security_application_user_set.security_users_id
        from
            security_application_user_set
        where
            security_application_user_set.security_users_id = ds.SECURITY_USERS_ID
    )
    or
    exists
    (
        select
            sru.role_user_id
        from
            dbo.security_role_user sru,
            dbo.security_department_role sdr,
            dbo.security_role sr
        where
            sru.security_users_id = security_application_user_set.security_users_id
            and dbo.f_is_time_expired(sru.expiration_time) = 'N'
            and sdr.department_role_id not in(-1)
            and sdr.role_id = sru.role_id
            and sdr.department_id = sru.department_id
            and ds.department_role_id = sdr.department_role_id
            and sr.role_id = sdr.role_id
            and sr.security_application_id = security_application_user_set.application_id
    )
)
and ds.entity_id = ?
and ds.key_value = ?
order by ds.seq_no, ds.business_entity_subtype_id, ds.security_tag
```

Sample Module Security Script

```
with
  security_application_user_set as
  (
    select
       sau.application_user_id,
       sau.application_id,
       sau.security_users_id
    from
       dbo.security_application_user sau,
       dbo.security_application sa,
       dbo.security_users su
    where
       sa.internal_name = 'USER'
       and su.user_name = 'MANDY'
       and sau.application_id = sa.application_id
       and sau.security_users_id = su.security_users_id
       and sau.activated_flag = 'Y'
       and sau.suspended_flag = 'N'
       and su.account_active = 'Y'
       and su.account_locked_flag = 'N'
       and sa.active_flag = 'Y'
  )
  select
      ds.entity_id,
      ds.key_value,
      ds.DEPARTMENT_ROLE_ID,
      ds.seq_no,
      ds.business_entity_subtype_id,
      ds.security_tag,
      ds.permission_code,
      ds.application_id
  from
      web.data_security ds,
      security_application_user_set
  where
  (
      ds.APPLICATION_ID = security_application_user_set.application_id or ds.APPLICATION_ID is null
  )
  and
  (
      exists
      (
          select
              security_application_user_set.security_users_id
          from
              security_application_user_set
          where
              security_application_user_set.security_users_id = ds.SECURITY_USERS_ID
      )
      or
      exists
      (
          select
              sru.role_user_id
          from
              dbo.security_role_user sru,
              dbo.security_department_role sdr,
              dbo.security_role sr
          where
              sru.security_users_id = security_application_user_set.security_users_id
              and dbo.f_is_time_expired(sru.expiration_time) = 'N'
              and sdr.department_role_id not in(-1)
              and sdr.role_id = sru.role_id
              and sdr.department_id = sru.department_id
              and ds.department_role_id = sdr.department_role_id
              and sr.role_id = sdr.role_id
              and sr.security_application_id = security_application_user_set.application_id
      )
```

```
  )
  and ds.entity_id = pkg_benefit_s.f_get_entity_id('MODULES')
  and ds.key_value = (select md.module_id from dbo.modules md where md.module_code IN ('WORLD'))
  order by ds.seq_no, ds.business_entity_subtype_id, ds.security_tag
```

- Take a screenshot of the results and attach in the **Query Results** section of your JIRA.
- If you are returning **Security Tags** and security is not working properly for security tags please attach screenshot of returned tags in your local code in InitializeModuleSecuirtyTagsBR.executeAction. We will not be accepting the Jira without returned tags code debug screenshot

Navigation

If you think navigation permissions are being used or might be interfering with permission configurations then along with above Jira templates, you need to add additional section for navigation query.

- Open VTHREE-110155 - Getting issue details... STATUS and copy the description to your JIRA
  - To efficiently copy the template to your JIRA. While editing the JIRA details. On the bottom of the editable box, please click on the Text tab. Copy the text from the Template JIRA and place it to your own JIRA. Then click on the Visual tab to return to normal.
- Please populate the Query Values section which should include the fields to populate the query like security user id etc
- Please populate the issue description section describing what is not working property and is in conflict with the description of field security should work (described in this page).
- Please populate Technical details section with additional information about technical steps taken to reproduce the issue. This section can include Breakpoint Screenshots or Step description on how to reproduce the issue.
- Please execute the following query on your DB instance (Note that you have to fill the query values in place of question marks) and attach the screenshot of results

Navigation Query

```
WITH
sa as
(
    select sa.application_id from dbo.security_application sa where sa.internal_name = 'USER'
),
su AS
(
    select su.security_users_id from dbo.security_users su where su.user_name = 'MANDY'
),
nas as
(
    SELECT
        nas.navigation_security_id,
        nas.navigation_id,
        nas.permission_code,
        nas.security_users_id,
        nas.department_role_id,
        na.internal_name as navigation_name
    FROM
        web.navigation_security nas,
        web.navigation na
    WHERE
        nas.navigation_id = na.navigation_id
),
nas_role AS
(
    SELECT
        nas.navigation_security_id,
        nas.navigation_name,
        sru.priority,
        nas.permission_code
    FROM
        nas,
        dbo.security_department_role sdr,
        dbo.security_role_user sru,
        dbo.security_role sr,
        sa,
        su
    WHERE
        nas.department_role_id = sdr.department_role_id
        and sdr.department_id = sru.department_id
        and sdr.role_id = sru.role_id
```

```
        and sru.role_id = sr.role_id
        and sr.security_application_id = sa.application_id
        and sru.security_users_id = su.security_users_id
        and sdr.department_role_id not in(-1)
        and sru.start_date <= current_date
        and (select dbo.f_is_time_expired(sru.expiration_time) from dual) = 'N'
        and (select dbo.f_is_date_future_date(sru.start_date) from dual)= 'N'
),
nas_su AS
(
    SELECT
        nas.navigation_security_id,
        nas.navigation_name,
        -1 as priority,
        nas.permission_code
    FROM
        nas,
        su
    WHERE
        nas.security_users_id = su.security_users_id
),
combined_pcs AS
(
    select * from nas_su
    UNION
    select * from nas_role
)
select *
from combined_pcs
order BY combined_pcs.PRIORITY;
```

Related articles

- How to Configure Security Permissions
- Security
- Responding to Appscan Reports
- Container Managed Authentication
- Dynamic SecurityTags based on business data (via Client BR)